AI Large Model Types and Principles

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1. Course Content

- Understand and master the basic knowledge of AI large models and the types of AI large models used in AI embodied intelligence gameplay.

> **Tip**
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> - This lesson is a purely theoretical lesson. If you want to quickly get started and experience the functional gameplay, you can skip this lesson first.

2. Introduction to AI Large Models

An AI large model, namely a large-scale pre-trained model, is the product of the deep integration of big data, large computing power, and strong algorithms. Simply put, it is like an intelligent agent "fed" with massive knowledge and trained repeatedly. Through learning from a large amount of data, it grasps the patterns and regularities in the data, thereby possessing strong generality and generalization ability. This capability means that AI large models are no longer limited to a single task, but can flexibly apply the knowledge they have learned, like humans, to solve complex problems in multiple domains. Leveraging multi-source data such as Internet text and images, they learn data patterns, possess strong generalization ability, can adapt to various tasks through transfer learning or prompt engineering, and may also exhibit emergent capabilities such as logical reasoning and common-sense understanding that were not set before training.

- Open-source model community: ModelScope Community
- Alibaba Bailian Large Model: Bailian Large Model

3.Common AI Large Model Categories

3.1 Text Generation Large Model

- Based on the Transformer architecture, through unsupervised or supervised learning on massive text data, it learns the grammatical, semantic, and pragmatic rules of language, so that it can generate natural and fluent text according to the input prompt or context. It is pre-trained through unsupervised learning and is suitable for tasks such as text generation and dialogue systems; it converts all text tasks into a unified text-to-text problem, providing a

more flexible framework that can handle various tasks such as translation, summarization, and question answering.

- In content creation, it can automatically generate practical text such as articles, news, and comments, improving content production efficiency, and can also assist authors with creative ideation and text polishing.
- In the field of intelligent interaction, it can be applied to intelligent customer service and chatbots, generating natural and fluent replies to enhance the user experience.
- In personalized teaching, it can analyze questions, provide explanations of test points, solution ideas, and results, and can also help users learn languages.
- In machine translation, it can achieve automatic translation, and combined with speech models, it can also achieve simultaneous interpretation, everyday subtitle generation, and more.

3.1.1 Principle Introduction

Based on the Transformer architecture (especially the self-attention mechanism), through unsupervised/semi-supervised learning on massive text data, it models the probability distribution and semantic associations of language, achieving understanding and generation of natural language.

- Pre-training logic
  - Autoregressive (AR): such as the GPT series, which predicts the next token through a "causal language model" (e.g., "today's weather → today's weather is"), learning the contextual dependencies of text.
  - Autoencoding (AE): such as BERT, which predicts masked tokens through a "masked language model" (e.g., "today [mask] is sunny" → "the weather"), learning bidirectional contextual semantics.
- Key technologies
  - Attention mechanism: dynamically allocates the weights of different words in the text, capturing long-distance dependencies (e.g., "events mentioned earlier → impacts later").
  - Prompt Tuning: activates specific capabilities of the model through templates (e.g., "Please summarize the following content: {text}"), adapting to downstream tasks.
- Capability emergence
  With the expansion of parameter scale (such as the hundred-billion level), the model may exhibit capabilities such as logical reasoning, common-sense understanding, and few-shot learning that were not explicitly programmed during the pre-training stage.

3.2 Multimodal Large Model

It can process multiple types of input data, such as text, images, audio, and video. Through cross-modal learning, it understands the relationships between data of different modalities and integrates multimodal data together, so as to fully utilize the information of each modality and build a unified representation space, enabling data of different modalities to understand and combine with each other, and thereby perform more complex and intelligent tasks. It can be used for cross-modal retrieval, retrieving data of one modality based on data of another modality; in visual question answering, the model answers text questions based on image content; it can also perform image caption generation, generating natural language text that describes the image content; and it can realize multimodal dialogue, conducting dialogues involving multiple

modalities of information, with broad application prospects in complex environments such as healthcare, transportation, and security monitoring.

3.2.1 Principle Introduction

Through cross-modal alignment and joint modeling, it learns a unified representation space for data of different modalities, achieving semantic association and collaborative processing between modalities.

- Contrastive learning: such as the CLIP model, which maps the feature vectors of images and text into the same space, and trains through "image-text pair matching" (e.g., "a picture of a dog → the text 'dog'").
- Encoder-decoder architecture: such as DALL-E, where the text encoder extracts semantic features and the image decoder generates the corresponding image (text → image).

Fusion methods

- Early fusion: merges multimodal data at the input layer (e.g., concatenating text embeddings with image pixel features).
- Late fusion: processes the data of each modality separately, and fuses the results at the decision layer (e.g., first analyzing text sentiment and image color separately, then making a comprehensive judgment).

3.3 Speech Recognition Model

- Converts the input speech signal into textual information. It is usually based on deep learning algorithms: first extract features from the speech signal, then input the features into a neural network model for training and recognition, and the model learns from a large amount of speech data to recognize different speech patterns and the corresponding textual content. It can help customer service personnel quickly record customer needs and problems, improving service quality and facilitating subsequent queries; it can be applied to voice search, freeing up the hands, and is suitable for various search environments such as vehicle navigation and mobile phones; it can also convert meeting conversations into text form, making it convenient to organize and record meeting content; in terms of human-computer interaction, it uses voice commands to control intelligent facilities, including hardware facilities such as robots and software applications.

  ### 3.3.1 Principle Introduction

  Converts the acoustic features of the speech signal into a text sequence, achieving end-to-end modeling based on deep learning.

  - Feature extraction: preprocesses the speech waveform (such as framing and windowing), extracting Mel-frequency cepstral coefficients (MFCC) or acoustic feature vectors (such as those extracted via CNN).
  - Sequence modeling: uses recurrent neural networks (RNN/LSTM) or a Transformer encoder to capture the temporal dependencies of the speech sequence (e.g., "continuous syllables → words").
  - Decoding mapping: through connectionist temporal classification (CTC) or attention mechanisms, maps the feature sequence to a text sequence (e.g., acoustic features → "hello").

3.4 Speech Synthesis Model

Converts the input text into a speech signal. Generally, a model is trained to learn the mapping relationship from text to speech; the model generates the corresponding speech features according to the input text, and then converts the features into audible speech through speech synthesis technology. It is widely used in fields such as voice assistants, audiobooks, and intelligent customer service, providing users with voice interaction services and enabling devices to communicate with users in natural and fluent speech.

3.4.1 Principle Introduction

Converts text semantics into natural and fluent speech signals, simulating the prosody, intonation, and emotion of human pronunciation.

Deep learning synthesis:

- Text analysis: parses the semantics, part of speech, and sentiment of the text through NLP models (e.g., "very happy today" → a cheerful tone).
- Acoustic modeling: uses the Tacotron series (encoder-decoder + attention mechanism) to generate the Mel spectrum of the speech.
- Vocoder: converts the Mel spectrum into a waveform signal, such as WaveNet and HiFi-GAN (improving the naturalness of the speech).

4.AI Model Comparison Summary

Model Type	Input	Output	Core Technology	Typical Scenario
Natural Language Large Model	Text	Text	Transformer self-attention	Writing, dialogue, translation
Multimodal Large Model	Text + image/audio	Cross-modal content	Cross-modal alignment, joint encoding	Image-text generation, visual question answering
Speech Recognition Model	Speech waveform	Text	Acoustic feature extraction + sequence decoding	Meeting transcription, voice search
Speech Synthesis Model	Text	Speech audio	Text analysis + acoustic modeling + vocoder	Voice assistants, audio content production